Daniel Jakab, Ronan Hwang
CS178
Midterm Summary Report
3/14/26

How we built the system

- Task Definition
    - Create a visual analytic system that plots students who were placed, did extracurriculars, or did placement training against their counterparts who either were placed or did not complete these. These plots have trend lines that help the user identify how different factors such as cumulative GPA, projects, internships, and soft skills correlate to each other for both of whichever of the previously described group was chosen.
- Interface Design
    - There are three drop-downs at the top of the screen, two for setting the x and y axis, and one for the facet which is the placement status, extracurricular status, or placement training status
    - Below this are the two scatter plots, one each for the facet being true or false. The scatter plots include a linear regression line which shows the correlation between what is on the x and y axis.
    - Below the scatter plots are the filters which allow us to exclude entries not in the ranges we select. The attributes we can filter upon are aptitude test score, soft skills rating, SSC marks, HSC marks, internships, and projects.
    - Finally, we have two selection boxes that allow us to filter out those who did or didn't do extracurriculars of placement training
- Parameterized SQL Query
    - SELECT "{x_col}", "{y_col}"
      FROM "placementdata.csv"
      WHERE {predicate} AND "{facet_col}" = '{fval}'
    - {predicate}="AptitudeTestScore" >= x AND "AptitudeTestScore" <= x
      AND "SoftSkillsRating" >= x AND "SoftSkillsRating" <= x
      AND "SSC_Marks" >= x AND "SSC_Marks" <= x
      AND "HSC_Marks" >= x AND "HSC_Marks" <= x
      AND "Internships" >= x AND "Internships" <= x
      AND "Projects" >= x AND "Projects" <= x
      AND "ExtracurricularActivities" IN (TRUE, FALSE)
      AND "PlacementTraining" IN (TRUE, FALSE)
      AND "PlacementStatus" = 'Placed'

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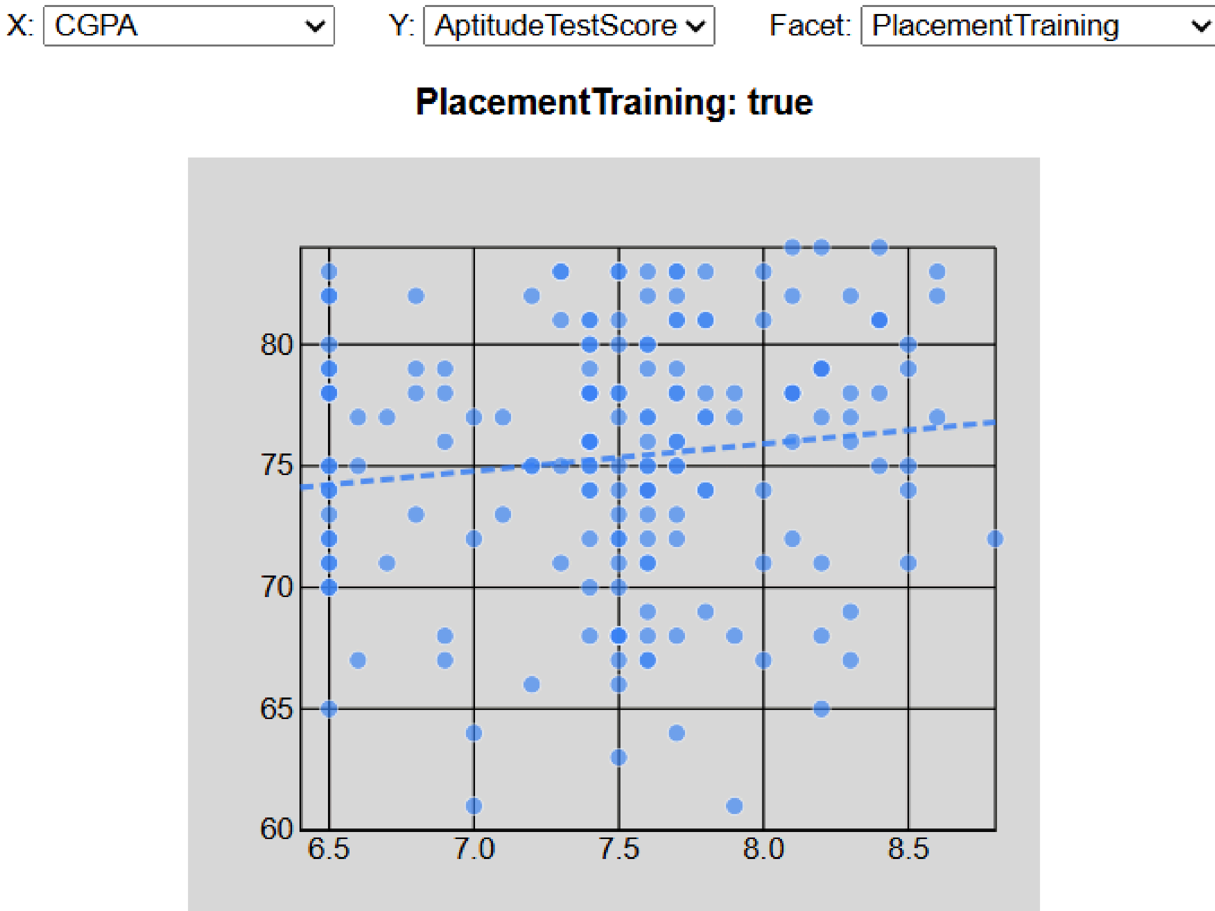
- How we used this to explore the dataset
    - By examining different axes, facets, and filters we are able to derive insights from our data. Several basic conclusions drawn from the plots are shown below



    - By plotting GPA against aptitude score, we can see that for those who were placed GPA and aptitude test score were positively correlated. This tells us that we can expect students with higher GPAs tend to have higher aptitude scores

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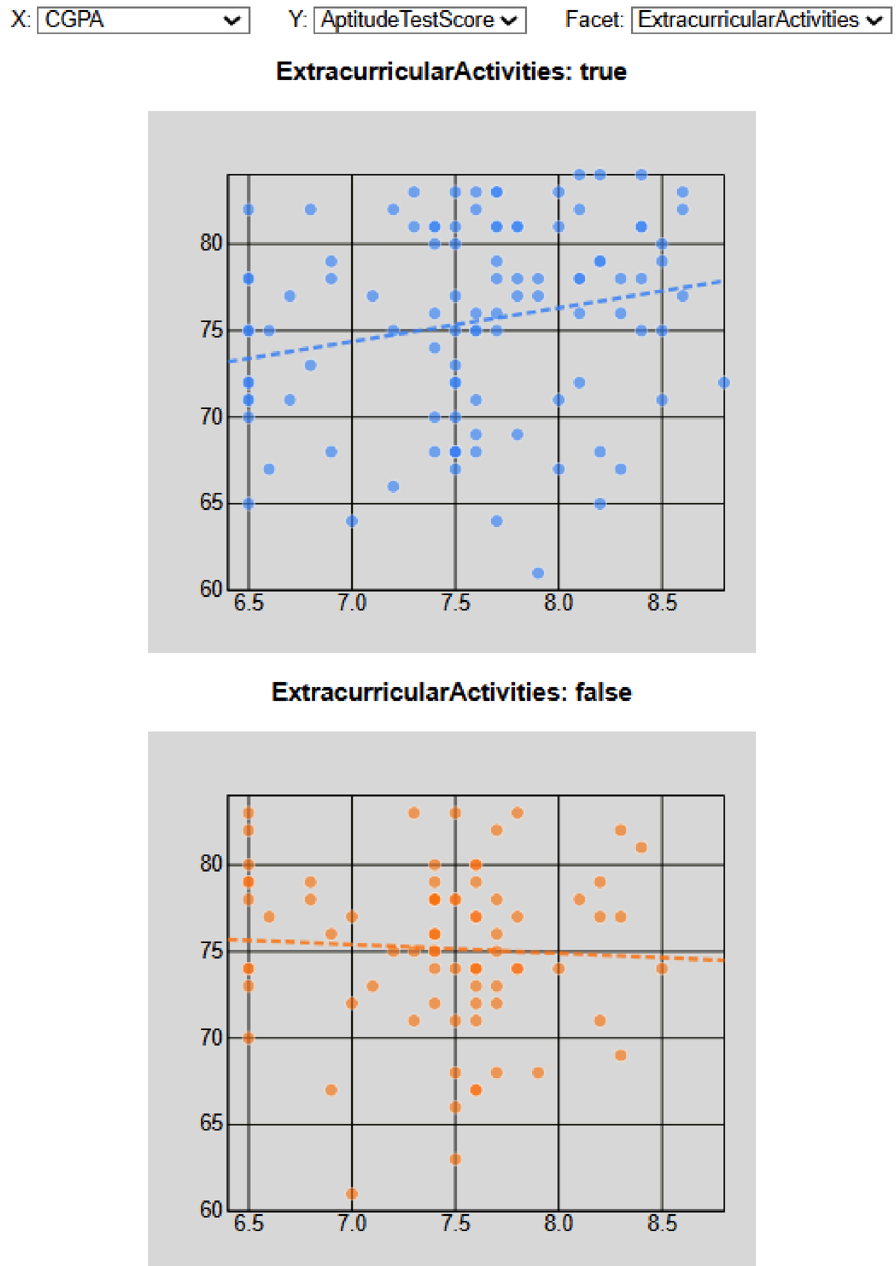

- When looking at extracurricular activities, we see this same correlation but only for those who did these extracurriculars.

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    For internships and projects, we see that for both students who did and didn't do placement training those who did more projects tended to have more internships.
- Key Insights
    - Aptitude Test Score is very correlated with being placed, more so than any of the other axis options
    - Extracurriculars are very correlated with being placed, especially for students who also did placement training

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- GPA and Aptitude Test Score correlate more for placed students than non placed students.